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Well Logging in Support for Seismic Reservoir Characterization

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SUMMARY

For the successful application of quantitative seismic interpretation techniques, namely the prediction of lithology, porosity and potentially pore fluid from seismic data, reliable calibration well data is key. Usually, the acquisition of a sophisticated logging suite involves higher costs, and we need hard facts for their legitimization. To prove the value of a modern logging suite, a pilot project over several wells in the Orenburg area has been conducted, where the results of domestic standard logging runs with extended measurements using foreign tools were compared. In this paper, we discuss both approaches and highlight what additional information we gain from the extended measurements. Using only the standard logging suite, it is very difficult to obtain accurate lithology, porosity and acoustic impedance values. For high quality reservoir characterization, in addition to the p-wave velocity, the shear wave and the Stoneley provide extremely useful information, especially for lithology estimation and the definition of permeable reservoir intervals. We show that in order to make use of the full potential of seismic inversion techniques, the complex of modern logging curves has to be acquired at least in a number of key wells for exploration and production.

Introduction

Over the last years, the industry has undertaken serious efforts to move from qualitative seismic interpretation – basically structural mapping and amplitude maps – to quantitative seismic interpretation, namely the prediction of lithology, porosity and potentially pore fluid from seismic data. The results help to mitigate the exploration risk and optimize the well pattern in the field development phase. However, for a successful prediction of the rock properties the access to reliable well data is key, to calibrate amplitude variations to physical effects of changes in the reservoir. Usually, the acquisition of a sophisticated logging suite involves higher costs, and we need hard facts for their legitimation. To prove the value of a modern logging suite, a pilot project over several wells in the Orenburg area has been conducted, where we compared the results of domestic standard logging runs with extended measurements using foreign tools. In this paper, we discuss both approaches and highlight what additional information we gain from the extended measurements.

Geological setting

The area under consideration is located on the south-eastern slope of Volga-Ural anticline within the Volga-Ural oil and gas bearing basin, European South-East. The sedimentary cover of the area is composed of Riphean siliciclastic graben fill, Middle Devonian to Lower Permian (Artinskian) mixed siliclastic-carbonate complexes, Kungurian sulfite-halogen complex and Upper Permian to Quaternary siliciclastics.

More than 20 reservoir intervals within the Middle Devonian - Lower Permian succession have been proved to be oil bearing and involved into development. Some gas discoveries were made in the Upper Permian Ufimian deposits. Major productive reservoirs vary from field to field, but as whole they are represented by Middle Devonian (D2) clastic and carbonate deposits (D7-D3), Upper Devonian (D3p) Pashian clastics (D0-D2), Frasnian-Famennian (D3f-fm, clinoforms of the so called Kalganski formation –Dkt), Frasnian to Tournaisian (D3f-C1t) carbonates including reefs (Df, Dzl, T1-T2), Visean (C1v) clastics (B2), Visean (C1v) Oksky carbonates (O6-O2).

Acquired well log data

Our usual logging program for an exploration or appraisal well in the Orenburg area so far comprised a set acquired with domestic tools, consisting of Acoustic log, Density log, Resistivity logs (lateral logging, micro-laterolog, contact logging, mud resistivity log), Conductivity logging (induction log), GR & Spectral Gamma-Ray, Neutron logging, Calliper and Spontaneous Potential (SP). The quality of the data was deemed to be sufficient for the evaluation of lithology, porosity and saturation of the reservoir section.

To tie well and seismic data correctly and make use of the advantages of simultaneous angle stack inversion, however, reliable shear information is required. So far there is no domestic tool for high quality shear data acquisition available - this represents a serious obstacle for utilizing the well data in quantitative interpretation of seismic data. Within the frame of a pilot program “Well logging in support for seismic”, in several wells over the section of interest in addition to the local logging standards a suite of modern western tools has been acquired by the same contractor. This program includes Cross-dipole acoustic measurements (P-wave velocity, S-wave velocity, Stoneley wave, azimuthal anisotropy as well as a complex including GR, Density, PEF, Neutron-porosity, and Calliper.

Comparison of Domestic and Western logging suites

The analysis of the data made clear that the domestic sonic log tool delivers a high quality p-wave velocity, meeting the requirements of quantitative seismic interpretation (Fig.1). But in view of the importance of Vp/Vs for lithology definition, the lack of a local tool for reliable shear wave measurements is a big disadvantage. Moreover, only shear waves measurements will give us indications of potential azimuthal anisotropy (fracture detection). In addition, without the Stoneley wave it is not possible to analyze the permeability intervals. The analysis of the extended pilot program revealed that the Stoneley derived permeability shows a very good match with the results of the well testing and core analysis, so it is considered as a very good tool to identify effective reservoir intervals. Therefore, for advanced reservoir characterization the necessity of using foreign equipment arises.

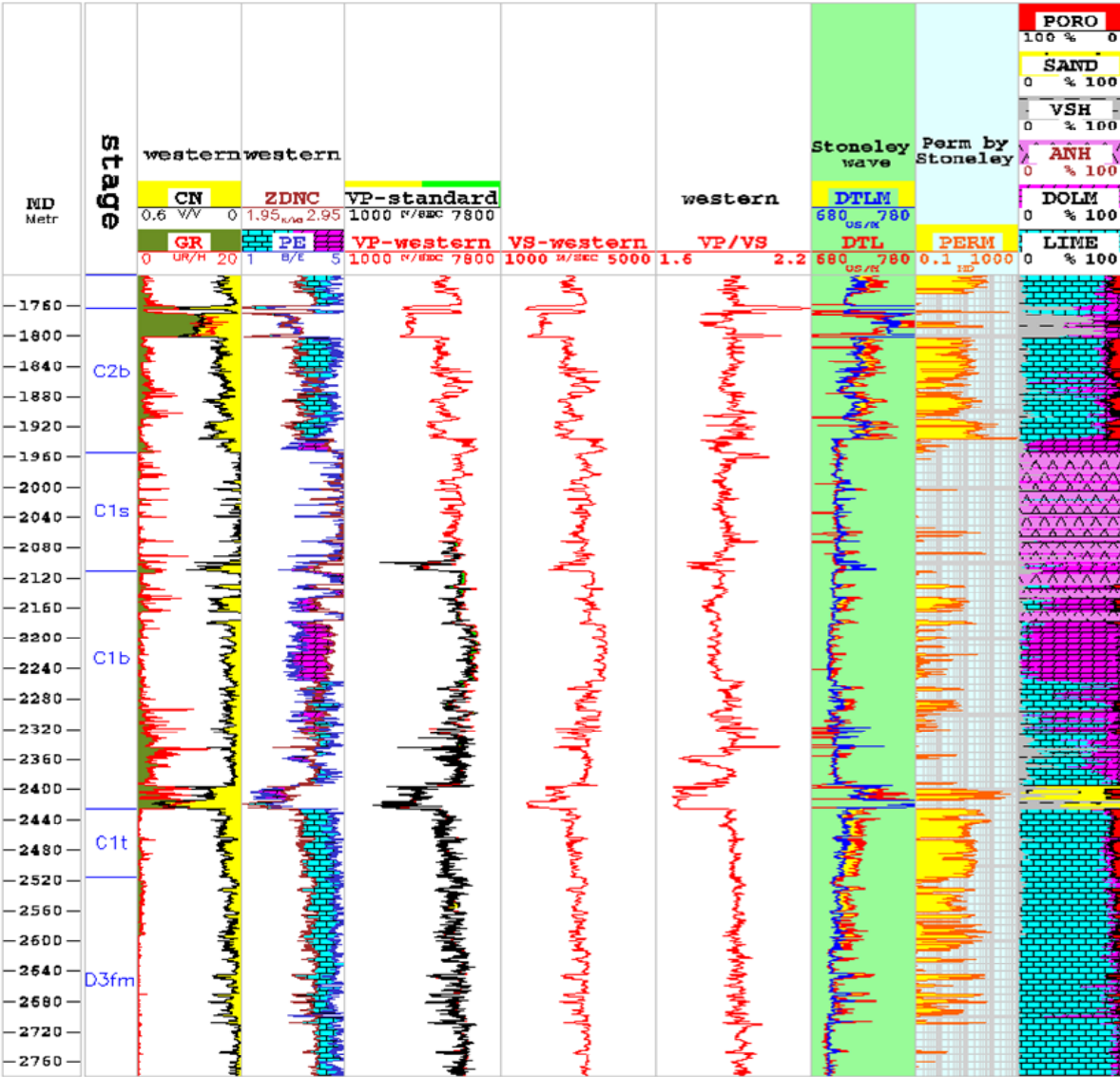


Figure 1 Example of a well showing the good match of p-wave for both, the domestic (“standard”) and foreign (“extended”) logging tools and the importance of advanced acoustic measurements for reservoir evaluation.

While analyzing, we observed a number of significant downsides for density/porosity measurements using the standard tools. The provided density logs are of poor quality and show a nonsystematic nonlinear shift along the borehole (Fig. 2). This is not only of relevance for the calculation of acoustic impedance as one of the key parameters derived from seismic data. Density is also one of the most important input parameters for porosity calculation, and if the porosity is wrong, the volumes and hence the reserves may not be correct. This critical issue led us to the conclusion that the standard density tool has to be exchanged for a modern, more reliable tool.

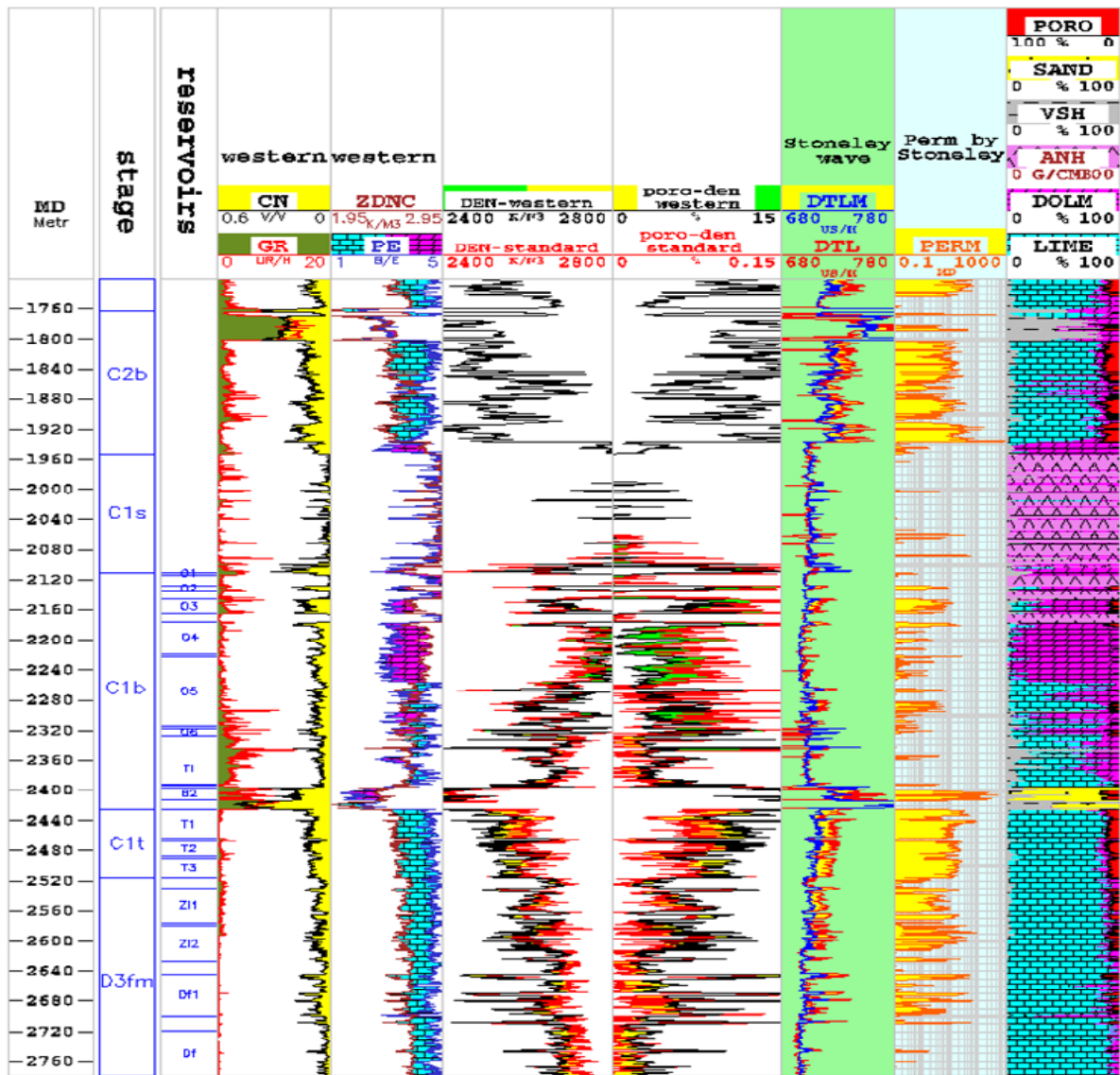


Figure 2 Example of differences in density measurements and porosity evaluation using domestic (“standard”) and foreign (“extended”) logging tools.

The analyzed standard data also suffer from the poor quality of the neutron log (Fig.3). Often, the Neutron log is still acquired in counts per second and needs to be transferred to porosity by the interpreter using empirical relationships. Much more accurate results are received, when the Neutron tool is already calibrated and the measurements are delivered in porosity units. The cross plots on figure 3 also demonstrate that the modern tools show a sharper separation by lithology and better correlation of porosity for both, limestones and dolomites. The standard tools exhibit a large scatter of values and hence, a larger uncertainty range.

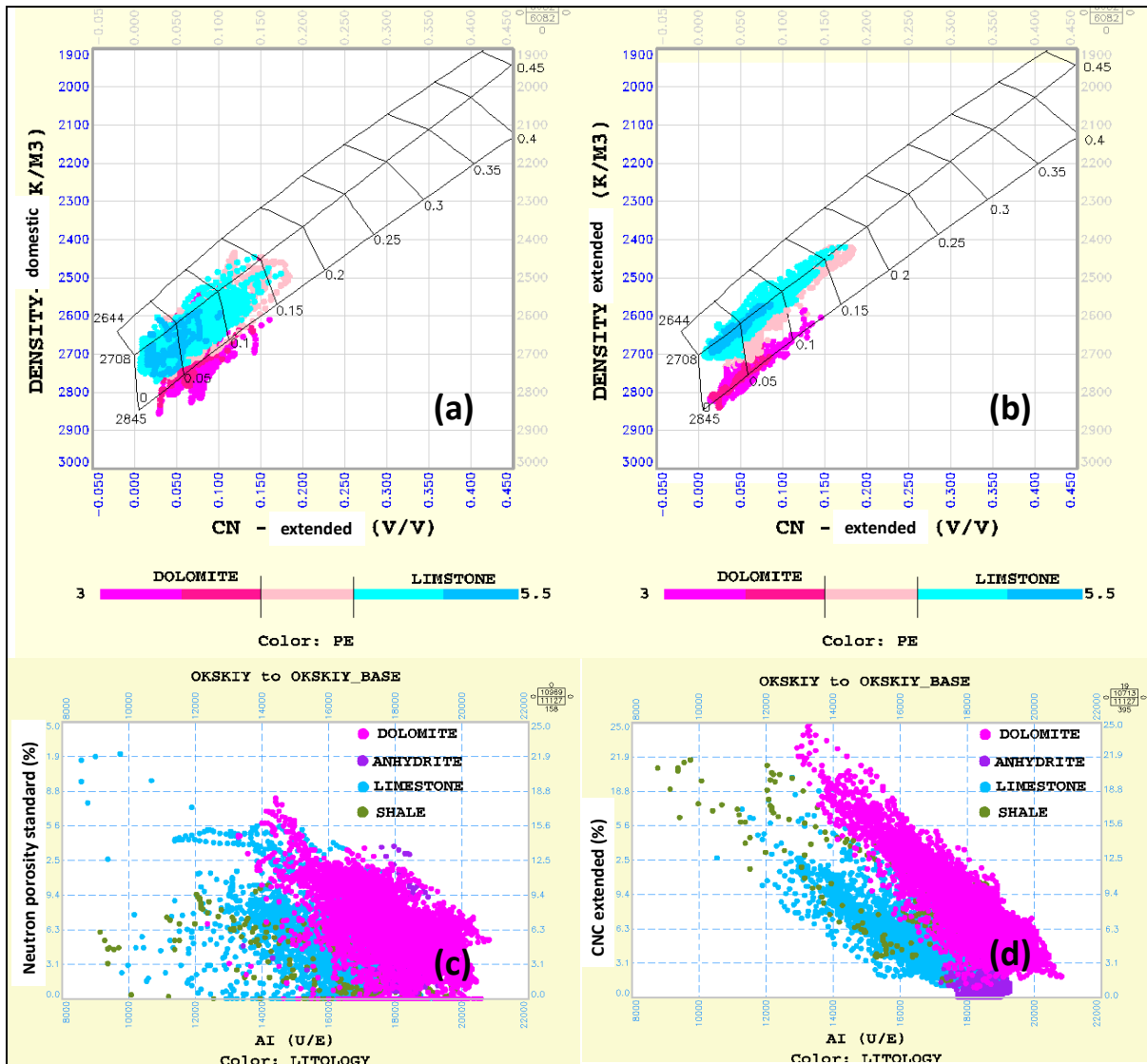


Figure 3 Cross plot analysis using five wells. Horizontal axis: measured data from "extended" logging suite (a and b – Calibrated Neutron, c and d – acoustic impedance), vertical axis: porosity data from "standard" tools (a and c) versus data from extended logging suite (b and d). Note the large uncertainty in lithology and porosity definition for the standard tools.

Conclusions

Using only the standard logging suite, it is rather difficult to obtain accurate lithology, porosity and acoustic impedance values. Intervals with fractured reservoirs may be overlooked. Especially the provided density measurements with older tools proved to be extremely unreliable, and this may have a direct impact of the reserves evaluation. In addition, the data are – due to the inherent blurring – applicable for property prediction from seismic data only in a qualitative manner. In view of the importance of the porosity estimation these standard tools have to be substituted by modern tools.

For high quality reservoir characterization, in addition to the p-wave velocity, the shear wave and the Stoneley provide extremely useful information, especially for lithology estimation and the definition of permeable reservoir intervals. To make use of the full potential of seismic inversion techniques, the complex of modern logging curves has to be acquired at least in a number of key wells for exploration and production.